i3 TECHNOLOGIES

i3 AI Skills & Assessment Cloud™

Professional Product, Architecture & Commercialization Blueprint

AI-Native Online Coding Tests • Certification Readiness • i3 AI-Lab • Admissions • Skills Intelligence • Digital Credentials • Talent

Strategic proposition

Build an AI-native skills verification platform that does more than test knowledge: it measures competence, observes practical performance in the i3 AI-Lab, identifies learning gaps, supports admissions, prepares learners for certification, issues verifiable credentials and creates employer-ready skills evidence.

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Source basis: the supplied 37-page coding-assessment market review, supplemented with current standards research on QTI 3.0, LTI Advantage, Open Badges 3.0 and secure sandboxing.

1. Executive Executive Summary

The supplied market review shows that leading assessment products compete on customizable challenges, candidate experience, language/framework coverage, scalability, reviewer collaboration, plagiarism controls, integrations and cost-effectiveness. It also highlights the limitations of overly intrusive assessment experiences and the value of realistic, appropriately scoped practical challenges. filecitetu rn0file0L40-L64

The strategic recommendation is therefore not to build another generic coding-test clone. i3 should build an integrated Skills Intelligence platform connecting assessment, practical AI-Lab execution, learning, certification readiness, admissions and verified credentials.

The platform should be designed as a multi-tenant SaaS product with an open integration layer and a proprietary i3 Skills Graph. QTI 3.0 should be used for portable assessment content; LTI Advantage should support LMS/tool interoperability; and Open Badges 3.0 should support verifiable digital credentials. 1EdTech states that QTI 3 supports portable assessment content, computer-adaptive testing and technology-enhanced interactions. citeturn0search0turn0search1

The resulting platform can be commercialized across four primary segments: individuals/certification candidates, universities and TVET institutions, enterprises/workforce development, and recruitment/talent assessment.

Recommended product family: i3 Assess™, i3 AI-Lab™, i3 CertReady™, i3 AdmitAI™, i3 Skills Passport™ and i3 Skills API™.

2. What the Supplied Market Review Tells Us

The supplied document identifies 30 coding/skills assessment products and repeatedly emphasizes flexibility, candidate experience, customizable challenges, language/framework support, applicant capacity, reviewer seats, scalability, support, plagiarism controls, pricing and time-to-hire. fileciteturn0file0L40-L64

Market capability	Examples identified in source	i3 opportunity
Real-world coding	Take-home / project-based assessments	Combine coding with live cloud/AI-Lab evidence
Language breadth	20–80+ languages/frameworks across vendors	Broad execution engine + prioritized enterprise stack
Screening	MCQ, coding, technical screening	Adaptive AI diagnostic + competency scoring
Interviewing	Live coding / pair programming	AI-assisted technical interview with human oversight
Integrity	Plagiarism/proctoring controls	Explainable, privacy-conscious assessment integrity
Customization	Own challenges, rubrics, integrations	AI-generated but human-approved assessment factory
Learning	Some platforms include practice/learning	Connect assessment directly to i3 AI-Lab and LMS
Certification	Some offer certification/proctored tests	Full readiness + remediation + evidence loop

The source also notes that candidate experience matters: excessively long or anxiety-inducing challenges and weak feedback can reduce the value of an assessment. It recommends clear evaluation criteria, sensible duration and an experience that enables candidates to demonstrate their ability. fileciteturn0file0L617-L642

3. Product Vision — i3 Skills Intelligence Platform

Position i3 as AI-powered Skills Verification Infrastructure, not merely an online test provider.

Layer	Purpose
i3 Assess™	Online diagnostics, coding, MCQ, simulations, projects, technical interviews and certification assessments
i3 AI-Lab™	Secure, ephemeral practical environments for coding, cloud, cyber, DevOps, AI/ML and enterprise technolog
i3 CertReady™	Certification readiness scoring, adaptive preparation, mock assessments and remediation
i3 AdmitAI™	Admissions diagnostics, programme-fit assessment and readiness recommendations
i3 Skills Passport™	Longitudinal, verifiable profile of skills, assessments, lab evidence and credentials
i3 Skills API™	APIs and webhooks for LMS, admissions, HR, ATS and partner integration
i3 Skills Graph	Core intelligence layer connecting competencies, questions, labs, learning resources, roles and credentials

4. Target User Journeys

4.1 Learner / Certification Candidate

Website → free diagnostic → AI skills profile → recommended pathway → assessment → AI-Lab → adaptive remediation → mock certification → readiness score → credential.

4.2 Admissions Applicant

Application → diagnostic assessment → academic/prior-learning evidence → practical assessment → AI-assisted interview → programme-fit score → human admissions decision → enrolment.

4.3 Enterprise

Define target role → AI generates competency model → assessment campaign → coding/practical lab → skills dashboard → gap analysis → learning recommendations → re-assessment → verified workforce skills.

4.4 Recruiter

Create role profile → assessment invitation → candidate completes technical assessment → skills evidence → integrity signals → human review → shortlist / interview.

Important: the platform should support human decision-making and explainable evidence rather than presenting AI scores as unquestionable admissions or hiring decisions.

5. Modern Feature Set — Recommended

Feature domain	Priority	Recommended capabilities
Assessment authoring	P0	AI assessment generator, templates, question bank, rubrics, versioning, randomization, approval w
Adaptive testing	P0	Difficulty adaptation, skill confidence, item selection, competency-based stopping
Coding engine	P0	Browser IDE, repositories, multi-file projects, unit tests, hidden tests, code execution, code review
AI Examiner	P0	Correctness, quality, security, complexity, testing, architecture and evidence-based feedback
AI Tutor	P0	Post-assessment remediation, hints, explanations, personalized practice
AI-Lab	P0	Ephemeral containers/VMs, cloud labs, enterprise technology scenarios, telemetry and evidence
Admissions integration	P0	Applicant sync, assessment launch, scores/status, programme recommendations
Certification readiness	P1	Readiness score, domain breakdown, mock exams, labs, adaptive remediation
Assessment integrity	P1	Code similarity, solution provenance, behavior anomalies, optional proctoring, human review

Feature domain	Priority	Recommended capabilities
AI Engineering mode	P1	Assess effective use of AI coding assistants with explicit policies and evidence
Digital credentials	P1	Open Badges 3.0, evidence, issuer, criteria, verification and revocation
Analytics	P1	Item analytics, competency trends, cohort analytics, predictive readiness
Marketplace	P2	Assessment/lab author marketplace, partner content and revenue sharing

6. Agentic AI Architecture

The platform should use a controlled multi-agent architecture rather than a single general-purpose chatbot.

Agent	Role	Human control
Assessment Designer Agent	Creates draft assessments, competencies, rubrics and variants	Required approval before publicat
Question Quality Agent	Checks ambiguity, correctness, difficulty, duplication and leakage risk	Human override
Assessment Examiner Agent	Grades objective and practical evidence	Explainable score + review queue
Code Review Agent	Reviews correctness, complexity, security, maintainability and testing	Rubric bounded
Lab Scenario Agent	Creates realistic practical incidents/projects	Template/rule validation
Proctor/Integrity Agent	Flags anomalies and evidence requiring review	Never final accusation
AI Interview Agent	Runs adaptive technical questioning	Interview rubric + human escalati
Tutor Agent	Generates remediation and practice	Policy controlled
Certification Agent	Calculates readiness and recommends next actions	Transparent model
Admissions Agent	Summarizes evidence and programme fit	Human admissions decision
Career Agent	Maps verified skills to roles and learning gaps	Recommendation only

AI-aware assessment modes

- Closed AI: no external AI assistance; suitable for controlled certification assessment.
- AI Allowed: AI tools permitted; assess verification, testing and engineering judgment.
- AI Engineering: explicitly assess the ability to use AI coding tools responsibly and effectively.

This avoids treating all AI use as cheating and creates a commercially relevant assessment category: AI-native engineering competence.

7. i3 Skills Graph — Core Intellectual Property

Every assessment item and practical lab should map to a structured competency model.

Object	Example
Domain	Cloud Engineering
Competency	Kubernetes Administration
Skill	Troubleshoot deployments
Sub-skill	Inspect pods, logs and deployment configuration
Evidence	Commands, changes, test result, explanation
Proficiency	Developing / Proficient / Advanced

Object	Example
Learning resource	Course, module, lab or remediation activity
Credential	i3 verified competency / badge
Role mapping	Cloud Engineer / DevOps Engineer

This creates a longitudinal data model in which assessment, learning and practical performance reinforce one another.

8. i3 AI-Lab Integration

The AI-Lab should become the platform's evidence engine. Instead of assessing only whether a candidate can answer a question, it can observe whether they can actually perform a realistic task.

Lab capability	Example evidence
Coding sandbox	Compilation, tests, code evolution, complexity
Linux	Commands, configuration, troubleshooting path
Docker	Images, containers, networking, security settings
Kubernetes	Deployments, services, logs, troubleshooting, RBAC
Cloud	Resource design, deployment, cost/security decisions
Cybersecurity	Detection, investigation, configuration and remediation
AI/ML	Model/API implementation, evaluation, safety checks
Enterprise technology	IBM / Red Hat / Microsoft / AWS / Google scenarios

For untrusted or AI-generated code, use strong isolation. gVisor documents sandboxing for untrusted code and Kubernetes integration, while noting performance/security trade-offs that should be evaluated during production engineering. citeturn0search12turn0search13turn0search14

9. Reference Technical Architecture

Recommended architecture:

Layer	Recommended responsibility
Experience	React/Next.js web application, responsive candidate UI, admin console
Identity	OIDC/OAuth2, RBAC/ABAC, tenant isolation, MFA for privileged users
API	API Gateway + versioned REST/GraphQL interfaces + webhooks
Assessment services	Assessment authoring, delivery, scoring, adaptive engine, scheduling
Code execution	Queue → execution orchestrator → isolated sandbox → test runner → evidence store
AI Gateway	Model routing, prompt/version management, policy controls, evaluation and cost governance
Agent runtime	Tool-using agents with scoped permissions, audit trails and approval gates
Skills Graph	Graph/relational hybrid for competencies, evidence, mappings and recommendations
Data	PostgreSQL + object storage + search + event stream + analytics warehouse
AI-Lab	Kubernetes-based ephemeral environments with sandboxing and lifecycle controls
Integration	LTI Advantage, QTI, APIs, webhooks, admissions, LMS, ATS/HRIS
Credentials	Open Badges 3.0 / verifiable credential issuance and verification

Layer	Recommended responsibility
Security	WAF, secrets management, encryption, tenant isolation, audit logging, SIEM integration

QTI 3.0 is a strong interoperability foundation because it is designed to exchange assessment items/tests and results across authoring tools, item banks, learning platforms, delivery and analytics systems; QTI 3 also natively supports computer-adaptive testing and portable custom interactions. citeturn0search0turn0search5

LTI Advantage is appropriate for LMS integration because it supports secure exchange of users/roles, assignments/scores and deep-linked content between learning platforms and external tools. citeturn0search10

10. Security, Privacy & Assessment Integrity

- Run every arbitrary code submission in a disposable, isolated execution environment.
- Apply CPU, memory, disk, process, network and execution-time limits.
- Separate candidate data, code, lab telemetry and credentials by tenant.
- Encrypt data in transit and at rest; manage secrets centrally.
- Maintain immutable audit trails for assessment creation, publication, grading and credential issuance.
- Use role-based and attribute-based access controls for administrators, reviewers, instructors and employers.
- Design integrity signals as explainable evidence. Flag for human review rather than automatically declaring misconduct.
- Support configurable data retention and deletion policies.
- Plan for privacy-by-design and regional data-residency requirements as the platform expands across Africa and globally.

11. Digital Credentials & Skills Passport

Use Open Badges 3.0 rather than proprietary PDF certificates as the strategic credential layer. 1EdTech describes Open Badges as information-rich, verifiable achievements that can include issuer, criteria and evidence; the current implementation guidance states that Open Badges 3.0 credentials are digitally signed and compatible with the W3C Verifiable Credentials data model. citeturn0search2turn0search8

Credential field	Example
Achievement	i3 Certified AI-Ready Cloud Engineer
Skills	Cloud, Linux, Python, Kubernetes, AI Engineering
Evidence	Assessment + AI-Lab practical + project
Issuer	i3 Technologies
Date	Issue / expiry where appropriate
Verification	Public verification endpoint
Alignment	Certification / competency framework

The i3 Skills Passport should aggregate verified achievements over time, enabling learners and professionals to share evidence with institutions and employers.

12. Admissions Application Integration

Admissions should be a first-class workflow rather than an afterthought.

Stage	System behavior
Application	Create/identify applicant and programme
Diagnostic	Assign programme-specific assessment
Evidence	Collect scores, prior learning and practical evidence
AI summary	Generate explainable applicant readiness summary
Programme fit	Recommend foundation, standard or advanced pathway
Human decision	Admissions team approves/rejects/requests more evidence
Enrolment	Push decision and onboarding status back to admissions
Learning handoff	Create learner profile and personalized learning plan

Strategic benefit: the assessment becomes a conversion and placement engine for i3, not simply an examination tool.

13. Commercialization Strategy

Product	Customer	Indicative model
Free Skills Diagnostic	Individuals / prospects	Free; lead generation
i3 CertReady	Certification candidates	\$20–\$100 per preparation package
i3 Skills Passport	Individuals	\$5–\$15/month premium tier
i3 Assess Campus	Universities / TVET	Annual institutional license by learner volume
i3 Workforce	Corporates	Per employee/year or enterprise license
i3 Talent Assess	Recruiters / employers	Per assessment or subscription
i3 AI-Lab	Training providers / enterprises	Per learner/hour, credits or annual capacity
White Label	Institutions / partners	Annual platform + implementation
i3 Skills API	Technology partners	Usage-based API pricing
Assessment Marketplace	Content authors / partners	Revenue share

Recommended go-to-market sequence

- 1. Existing i3 ecosystem: admissions + training + certification preparation.
- 2. Universities/TVET: campus assessment and practical labs.
- 3. Enterprise workforce: skills inventory and reskilling.
- 4. Recruitment: technical assessment and verified skills evidence.
- 5. Platform/API: white-label and ecosystem expansion.

The first commercial wedge should therefore be Admissions + Certification Readiness + AI-Lab, where i3 can leverage existing relationships and capabilities rather than entering the recruitment market cold.

14. Indicative Pricing Architecture

Tier	Indicative starting point	Purpose
Free	\$0	Diagnostic + basic profile + limited practice
Learner Pro	\$8–\$15/month	Adaptive assessments + AI Tutor + Skills Passport
CertReady Pack	\$39–\$99	Certification diagnostic + mocks + remediation + labs

Tier	Indicative starting point	Purpose
Campus	$3–$12/active learner/year	High-volume assessment; final price based on scope
Enterprise	$10–$40/employee/year	Skills assessment and workforce intelligence
Talent	$10–$50/assessment	Technical screening / assessment evidence
AI-Lab	Credits / usage	Practical environment consumption
White Label	$10k–$50k+ implementation	Branding, integrations, dedicated tenancy
Enterprise Platform	Custom	SLA, private deployment, integrations, data residency

These are planning ranges, not market quotations. Final pricing should be validated through 3–5 pilot customers and calibrated to compute, AI-model, lab-runtime and support costs.

15. 18-Month Product Roadmap

Phase	Timeline	Major deliverables
Foundation	0–3 months	Architecture, identity, tenants, assessment builder, question bank, basic coding engine, admissions API
MVP	3–6 months	AI question generation, AI grading, coding sandbox, dashboards, skills graph v1, i3 AI-Lab integration
Scale	6–9 months	Adaptive testing, certification readiness, integrity analytics, QTI import/export, LTI integration
Credential	9–12 months	Open Badges 3.0, Skills Passport, AI Interviewer, enterprise dashboards
Platform	12–15 months	Workforce skills intelligence, white label, public API, partner integrations
Ecosystem	15–18 months	Assessment marketplace, employer/talent workflows, multilingual expansion, regional deployment options

16. Business & Product KPIs

Category	KPIs
Growth	Free diagnostics, conversion to paid, institutional pipeline, CAC
Assessment	Completion rate, median completion time, item discrimination, question leakage
Learning	Skill improvement, remediation completion, repeat assessment improvement
AI	Grading agreement with experts, hallucination/error rate, cost per assessment, agent escalation rate
Lab	Lab launch success, runtime utilization, failure rate, cost per learner
Admissions	Assessment completion, application-to-enrolment conversion, time-to-decision
Credentials	Credentials issued, verified, shared and employer viewed
Enterprise	Employees assessed, skills gaps closed, renewal rate
Platform	API calls, tenants, active users, gross margin, uptime

17. Build vs Buy Strategy

Capability	Recommendation	Reason
Assessment UX	BUILD	Core product differentiation
Skills Graph	BUILD	Strategic IP
AI agent orchestration	BUILD + MODEL-AGNOSTIC	Avoid vendor lock-in
Code execution	BUILD/INTEGRATE	Security and scale are core

Capability	Recommendation	Reason
AI-Lab	BUILD around existing i3 assets	Major differentiator
Admissions integration	BUILD	Existing i3 workflow
LMS integration	INTEGRATE	Avoid duplicating Skillsoft/LMS functionality
QTI/LTI	IMPLEMENT STANDARD	Interoperability
Credentials	IMPLEMENT STANDARD	Trust and portability
LLM models	MULTI-MODEL	Cost, quality, resilience and regional options

The key principle is: own the intelligence, workflow and evidence model; integrate commoditized infrastructure where appropriate.

18. Competitive Differentiation

Conventional assessment	i3 target state
Question → score	Assessment → evidence → skills graph
Coding only	Coding + cloud + AI-Lab + simulation
Static test	Adaptive competency assessment
Generic feedback	AI-personalized remediation
Certificate PDF	Verifiable digital credential + evidence
Recruitment only	Admissions + learning + certification + workforce
AI as chatbot	Controlled multi-agent assessment system
Proctoring as surveillance	Explainable assessment-integrity evidence
Vendor-specific content	QTI-based portable content

19. Key Risks & Mitigations

Risk	Mitigation
AI grading inconsistency	Rubrics, benchmark sets, expert calibration, model evaluation, human escalation
Question leakage	AI-generated variants, randomized pools, secure item handling, exposure analytics
Code sandbox escape	Strong isolation, network controls, quotas, ephemeral environments, security testing
AI cheating	Explicit assessment modes; assess AI collaboration where appropriate
Privacy concerns	Data minimization, configurable retention, transparent integrity policies
High AI/lab costs	Usage quotas, model routing, caching, credits and workload optimization
Scope creep	Start with admissions + certification + AI-Lab MVP
Weak differentiation	Invest early in Skills Graph, practical evidence and Skills Passport

20. Recommended Initial Product Team

Role	Initial focus
Product/Domain Architect	Overall platform, standards and roadmap
Full-Stack Engineers	Candidate/admin applications + APIs

Role	Initial focus
Platform/Cloud Engineer	Kubernetes, execution and AI-Lab orchestration
AI/ML Engineer	Agents, grading, adaptive models and evaluation
Assessment Specialist	Psychometrics, rubrics and question quality
Security Engineer	Sandbox, IAM, privacy, threat modeling
UX/Product Designer	Candidate and instructor experience
QA/Automation	Assessment correctness, load, security and regression
DevOps/SRE	CI/CD, observability, reliability and cost

21. First 90 Days

Days	Outcome
0–30	Finalize product requirements, competency model, UX prototype, architecture, security model and admissions API contract
31–60	Build assessment authoring, candidate delivery, coding execution MVP, skills model and AI question-generation prototype
61–90	Connect i3 AI-Lab, launch pilot assessment, implement reporting, admissions handoff and expert validation

Target pilot: one i3 certification-preparation programme and one admissions workflow, with a controlled cohort and expert-graded benchmark set.

22. Final Strategic Recommendation

i3 should build this as a platform business, not as a feature inside the LMS or admissions application.

The highest-value architecture is:

i3 Assess → i3 AI-Lab → i3 Skills Graph → i3 CertReady / AdmitAI → i3 Skills Passport → i3 Workforce / Talent

The strongest moat will not be the coding editor. It will be the combination of verified practical evidence + longitudinal skills intelligence + agentic assessment + admissions/learning integration + portable credentials.

The supplied market review validates the need for candidate-friendly, customizable, scalable assessment technology, while current interoperability standards provide a practical route to making i3 portable across LMS and assessment ecosystems. fileciteturn0file0L627-L642 QTI 3.0 provides the assessment portability foundation, LTI Advantage supports learning-tool integration, and Open Badges 3.0 provides a modern verifiable-credential layer. citeturn0search0turn0search10turn0search8

The strategic objective should therefore be to make i3 the AI-native skills verification and practical assessment platform for Africa, with global interoperability from day one.

23. Key References

1. Supplied source document: Discover the 30 best coding assessment tools that will help you identify and hire the right candidates and build a world-class technical team. Source-derived findings are cited inline. fileciteturn0file0L1-L8

2. 1EdTech QTI Specification Documents and QTI 3.0 overview. citeturn0search0turn0search5

3. 1EdTech LTI Advantage guidance. citeturn0search10

4. 1EdTech Open Badges 3.0 and implementation guidance. citeturn0search2turn0search8

5. gVisor security and Kubernetes sandboxing documentation. citeturn0search12turn0search13turn0search14

Note: commercial pricing, roadmap durations, architecture choices and product recommendations in this document are strategic recommendations rather than claims directly established by the supplied source.